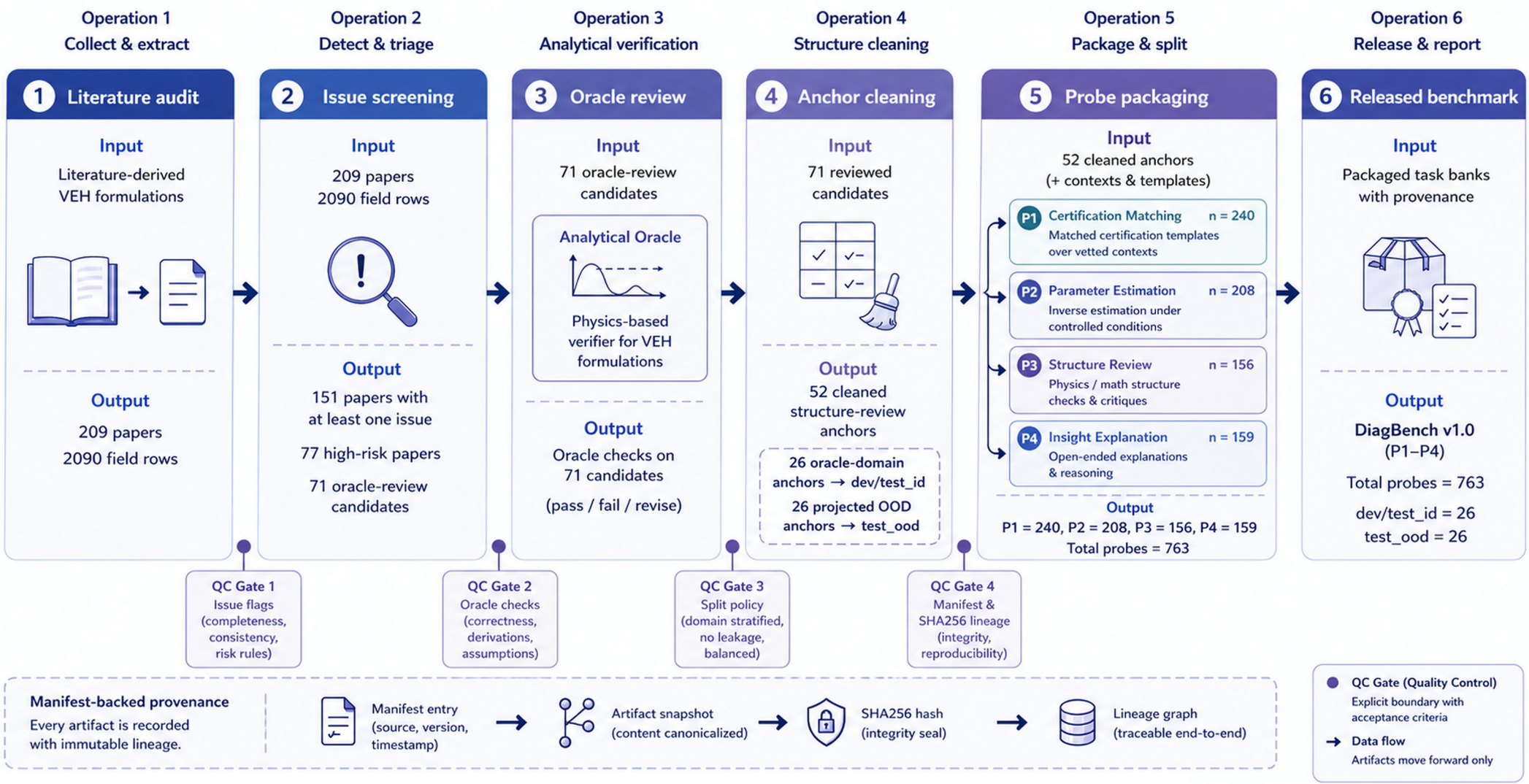
Operation 1
Collect & extract
Operation 2
Detect & triage
Operation 3
Analytical verification
Operation 4
Structure cleaning
Operation 5
Package & split
Operation 6
Release & report
1 Literature audit
Input
Literature-derived VEH formulations
Output
209 papers
2090 field rows
2 Issue screening
Input
209 papers
2090 field rows
Output
151 papers with at least one issue
77 high-risk papers
71 oracle-review candidates
3 Oracle review
Input
71 oracle-review candidates
Analytical Oracle
Physics-based verifier for VEH formulations
Output
Oracle checks on 71 candidates
(pass / fail / revise)
4 Anchor cleaning
Input
71 reviewed candidates
Output
52 cleaned structure-review anchors
26 oracle-domain anchors → dev/test_id
26 projected OOD anchors → test_ood
5 Probe packaging
Input
52 cleaned anchors
(+ contexts & templates)
P1 Certification Matching n = 240
Matched certification templates over vetted contexts
P2 Parameter Estimation n = 208
Inverse estimation under controlled conditions
P3 Structure Review n = 156
Physics / math structure checks & critiques
P4 Insight Explanation n = 159
Open-ended explanations & reasoning
Output
P1 = 240, P2 = 208, P3 = 156, P4 = 159
Total probes = 763
6 Released benchmark
Input
Packaged task banks with provenance
Output
DiagBench v1.0
(P1–P4)
Total probes = 763
dev/test_id = 26
test_ood = 26
QC Gate 1
Issue flags (completeness, consistency, risk rules)
QC Gate 2
Oracle checks (correctness, derivations, assumptions)
QC Gate 3
Split policy (domain stratified, no leakage, balanced)
QC Gate 4
Manifest & SHA256 lineage (integrity, reproducibility)
Manifest-backed provenance
Every artifact is recorded with immutable lineage.
Manifest entry (source, version, timestamp)
Artifact snapshot (content canonicalized)
SHA256 hash (integrity seal)
Lineage graph (traceable end-to-end)
QC Gate (Quality Control)
Explicit boundary with acceptance criteria
Data flow
Artifacts move forward only